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WORLD ATHLETICS

KTH VETENSKAP OCH KONST

Advanced footwear technology and its performance enhancing effects: what is happening?

Is it cheating? Are there injury risks?





Toni Arndt

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Tigst Assefa broke the world marathon record in Berlin 2023 with a time of 2:11:53

By over a minute…

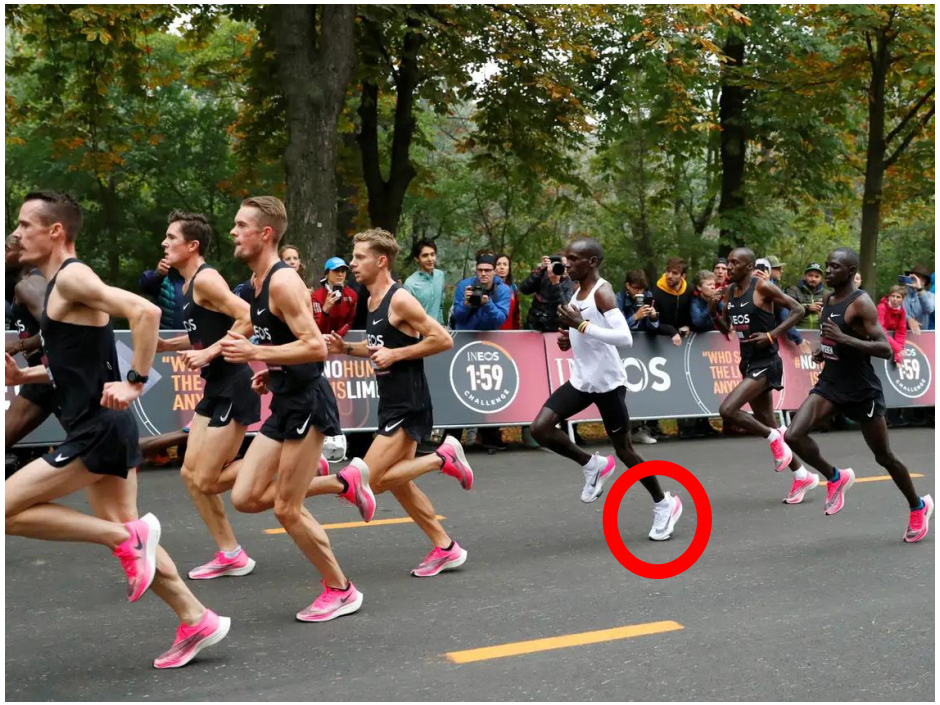


Ruth Chenpngetich broke the world marathon record in Chicago 2024 with a time of 2:09:56

By nearly another 2 minutes…

Both were wearing shoes with advanced footwear technology (AFT)

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In autumn 2019 Eliot Kipchoge ran a marathon distance in under 2 hours – a performance that was previously regarded as impossible.

In 2023 Kelvin Kiptum broke the marathon World Record with a time of 2:00:35 in the Chicago Marathon.

Both ran in Nike shoes with advanced footwear technology.

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Background

- Large international sports shoe companies have in recent years developed innovative running shoes which have been scientifically shown to improve performance
- Since the Olympic Games in Rio, 2016 running distances from 5,000 m to the marathon have been dominated by runners wearing such shoes
- A significant improvement in tack and field results has also been seen
- World Athletics now has a dilemma in regulating these shoes. But how? Is it cheating to run in them? Which shoes should be accepted?
- What are the mechanisms underlying the shoes' performance enhancing characteristics?

World Athletics technical rule 143, paragraph 5.2 which states 4 that:

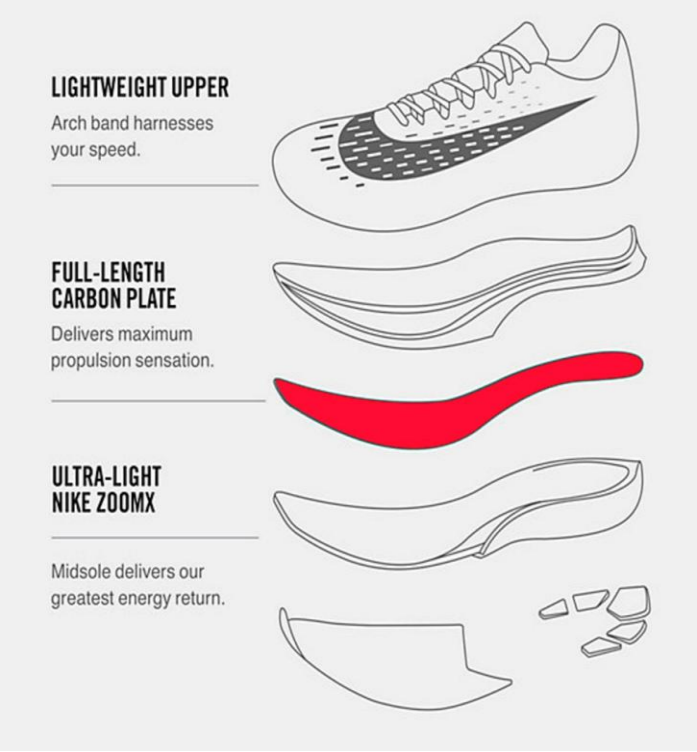
"…athletes may compete barefoot or with footwear on one or both feet. The purpose of shoes for competition is to give protection and stability to the feet and a firm grip on the ground. ***They must not give athletes any unfair assistance or advantage****. Any type of shoe must be reasonably available to all in the spirit of the universality of athletics…"*

Technological developments in sports – A question of fairness and integrity? Masters thesis GIH 2020 Frida Williamson.

Arndt *et al*, under review, Journal of Philosophy of Sports

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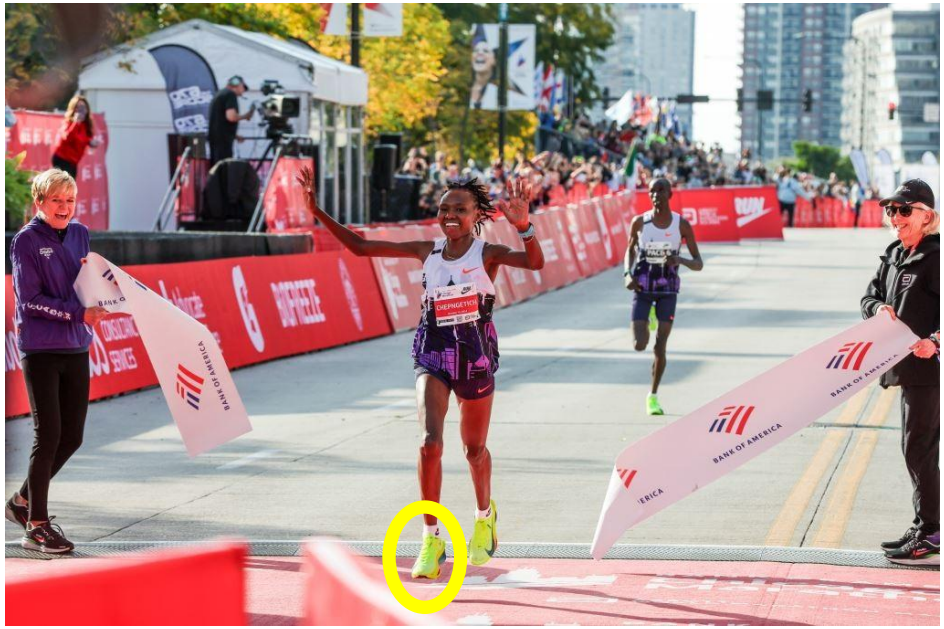
What are we talking about?





Nike's Latest Running Shoes Are So Good, They Might Be Too Good

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AFT performance enhancing characteristics

- Very low mass
- Sole material
- Sole height
- Longitudinal stiffness



- Combined effect of shoes and neuromusculoskeletal function

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The general strategy with stiff soles

- Muscles require less work for the same performance
- Decreased loss of energy at the metatarsophalangeal joint
- Decreased energy loss will have a measurable effect on running performance

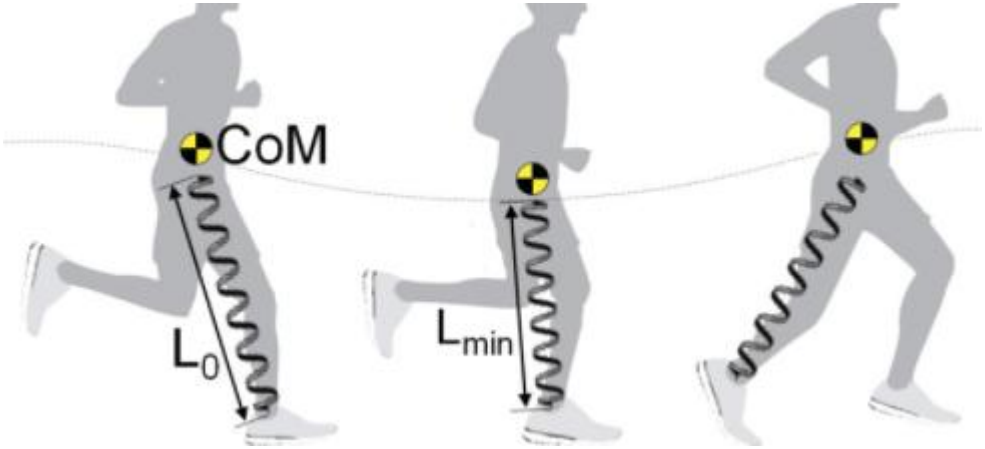


Stefanyshyn & Nigg, 2020

Tigst Assefa, Berlin 2023, 2:11:53

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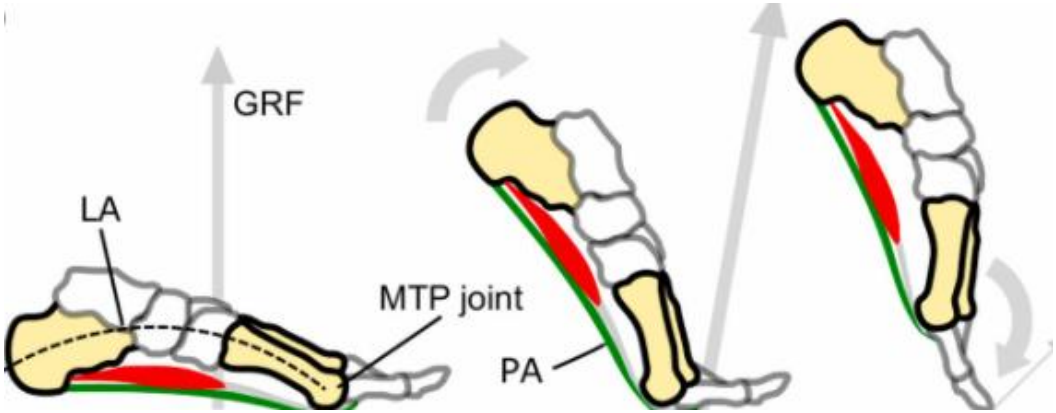
Sole compression



Energy stored in sole compression during loading can ideally be stored and released during unloading, i.e. the foam acts as a spring and releases energy during unloading

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Bending at forefoot



The metatarsophalangeal (MTP) joint is flexed in midstance and extends in late stance.

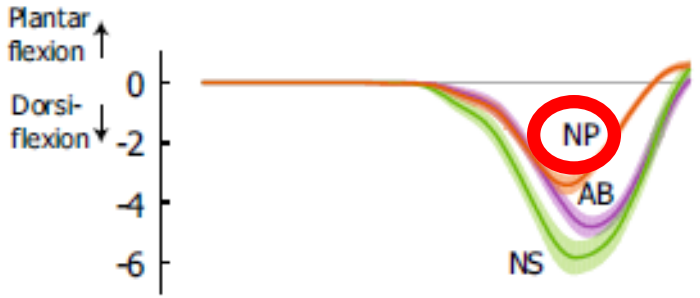
Stiff shoes reduce dorsiflexion and therefore the related energy loss.

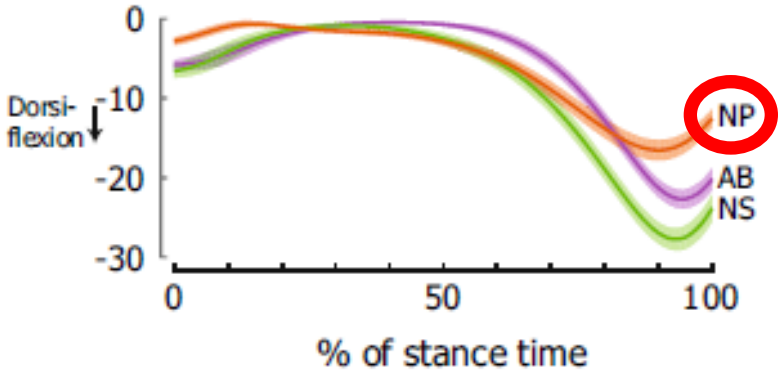
Roy & Stefanyshyn, 2006. Med Sci Sports Exerc.

Altered Biomechanics



- Metatarsophalangeal (MTP) joint
  - Lower negative work rate
  - Reduced dorsiflexion angle





Hoogkamer *et al*., 2018. Sports Medicine

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Longer lever arm due to stiff plate

1



Thick sole includes more energy returning material

2

3

4

5

New energy returning materials

Rigid element inserted in midsole

Stiffer MTP joint

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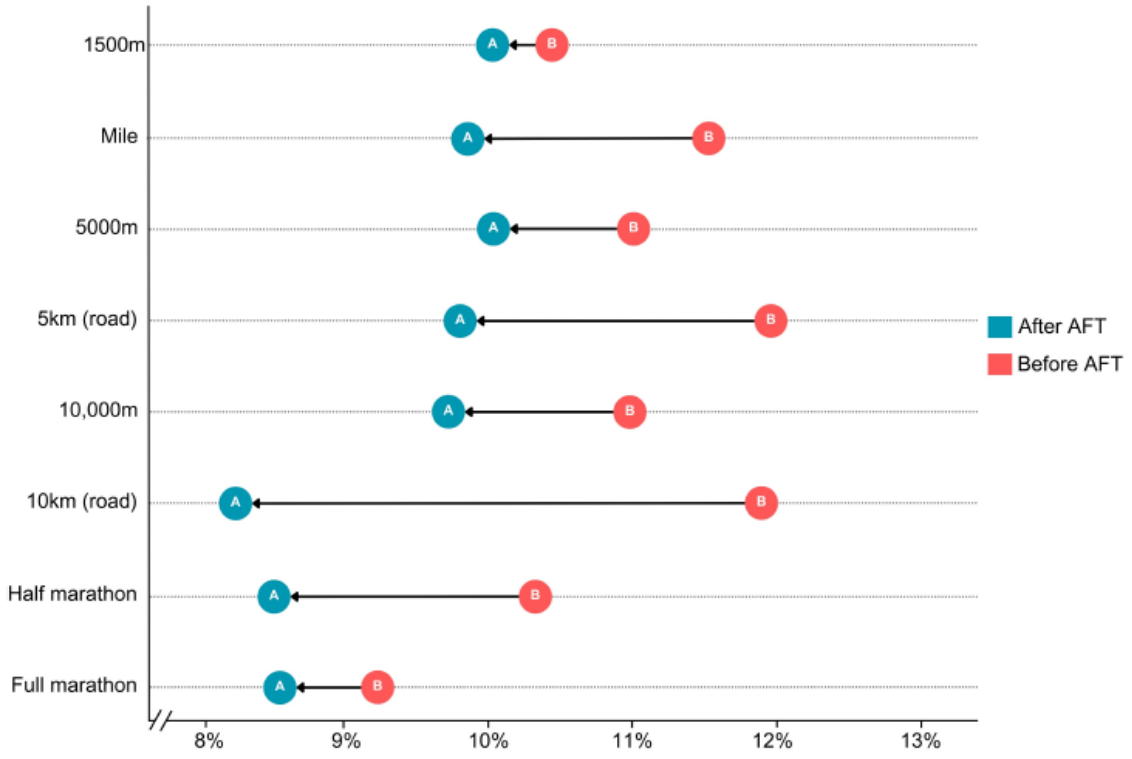
Evolution of top 20 male and female seasonal bests

	Year								
	2012	2013	2014	2015	2016	2017	2018	2019	F(7,152), p values
Male events									
10 km [min:s]	27:50.6 (00:08.9)	27:43.1 (00:07.4)	27:45.9 (00:11.3)	27:48.1 (00:07.0)	27:49.9 (00:9.6)	27:44.8 (00:16.1)	27:30.7 (00:14.5)	27:24.3 (00:16.4)	12.76, p<0.001
Half Marathon [h:min:s]	59:23.3 (00:18.7)	59:39.5 (00:27.0)	59:21.7 (00:15.1)	59:30.1 (00:10.6)	59:27.7 (00:15.3)	59:20.6 (00:17.0)	59:11.8 (00:18.2)	59:07.7 (00:20.2)	6.03, p<0.001
Marathon [h:min:s]	02:05:02.1 (00:28.7)	02:05:16.3 (00:58.7)	02:05:14.3 (01:03.2)	02:05:56.0 (00:38.7)	02:05:08.0 (01:06.2)	02:05:29.9 (00:49.4)	02:04:33.0 (00:51.8)	02:03:59.5 (00:57.6)	8.48, p<0.001
Female events									
10 km [min:s]	31:29.7 (00:21.4)	31:38.1 (00:28.5)	31:26.0 (00:19.9)	31:25.3 (00:25.9)	31:23.1 (00:23.9)	30:59.7 (00:28.9)	31:05.3 (00:21.8)	30:41.3 (00:24.4)	1.82, p<0.001
Half Marathon [h:min:s]	01:07:46.0 (00:33.0)	01:07:32.6 (00:42.1)	01:07:59.3 (00:53.9)	01:07:42.7 (01:00.0)	01:07:10.2 (00:45.8)	01:06:20.3 (00:51.2)	01:06:09.6 (00:43.7)	01:06:03.2 (00:23.4)	0.97, p<0.001
Marathon [h:min:s]	02:20:49.0 (01:10.5)	02:22:31.1 (01:13.9)	02:22:30.9 (01:35.4)	02:22:15.4 (01:27.1)	02:22:30.9 (01:12.3)	02:20:57.4 (01:26.1)	02:20:00.3 (01:13.6)	02:19:18.2 (01:38.3)	17.02, p<0.001

Bermon *et al.*, 2021. Frontiers in Sports and Active Living

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Sex gap in World records since introduction of AFT

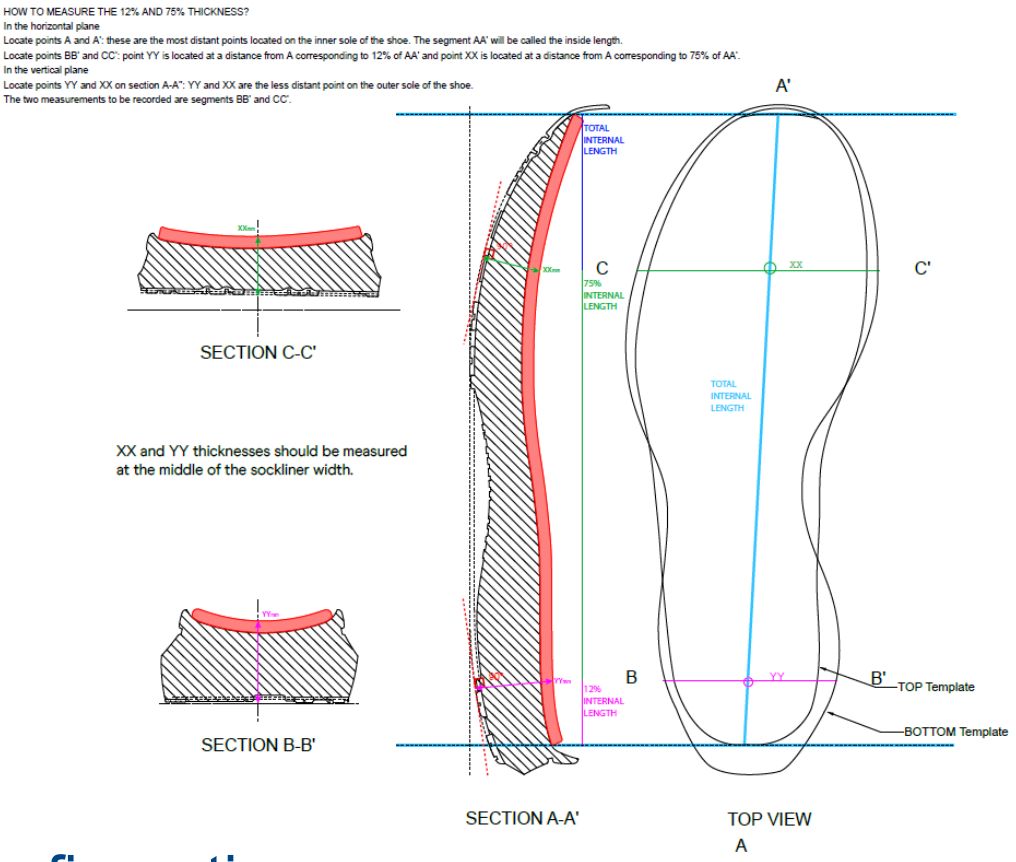


Mason *et al.*, 2024. Frontiers in Sports and Active Living

World Athletics rules (road shoes)

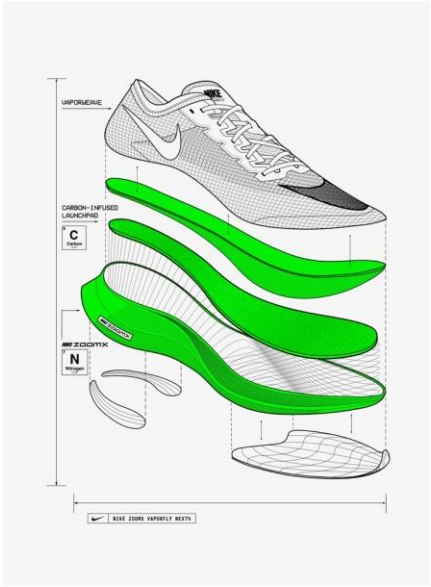
Regulate:

- Sole height



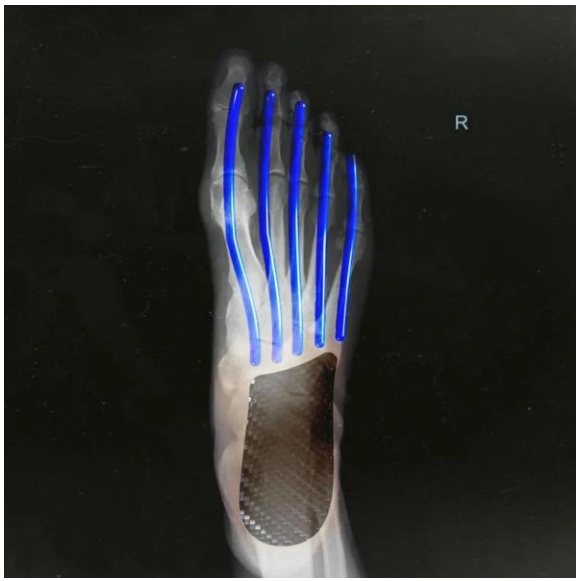
- Rigid element configuration

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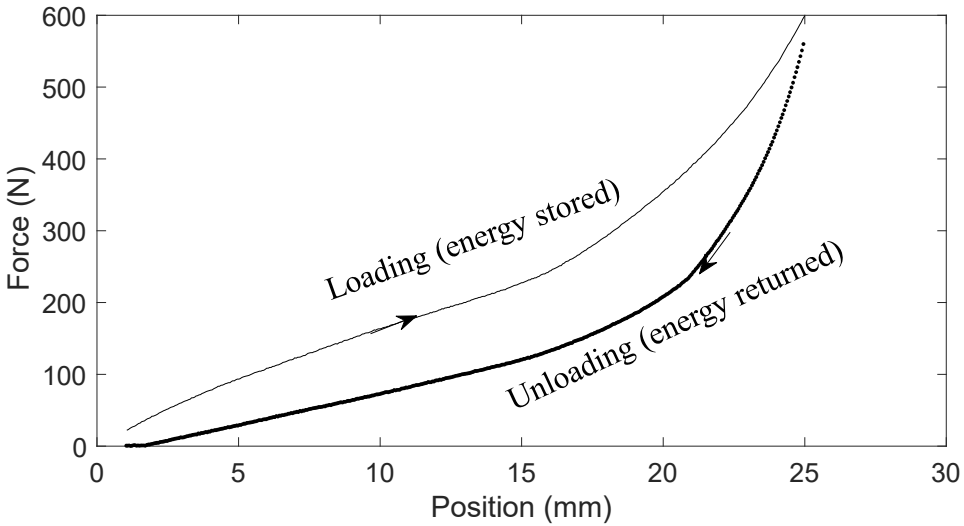


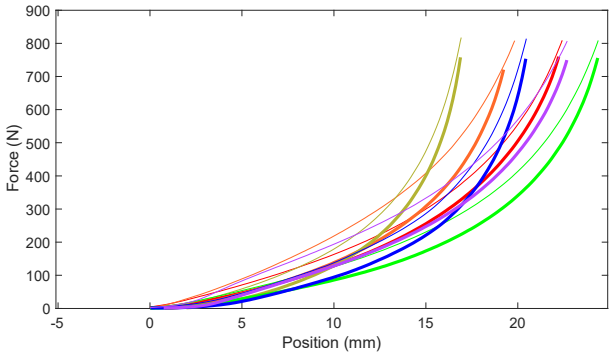


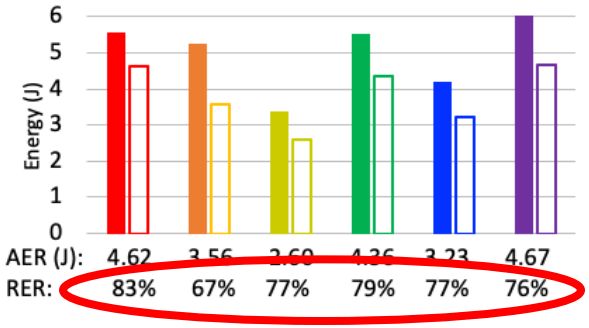
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Energy return

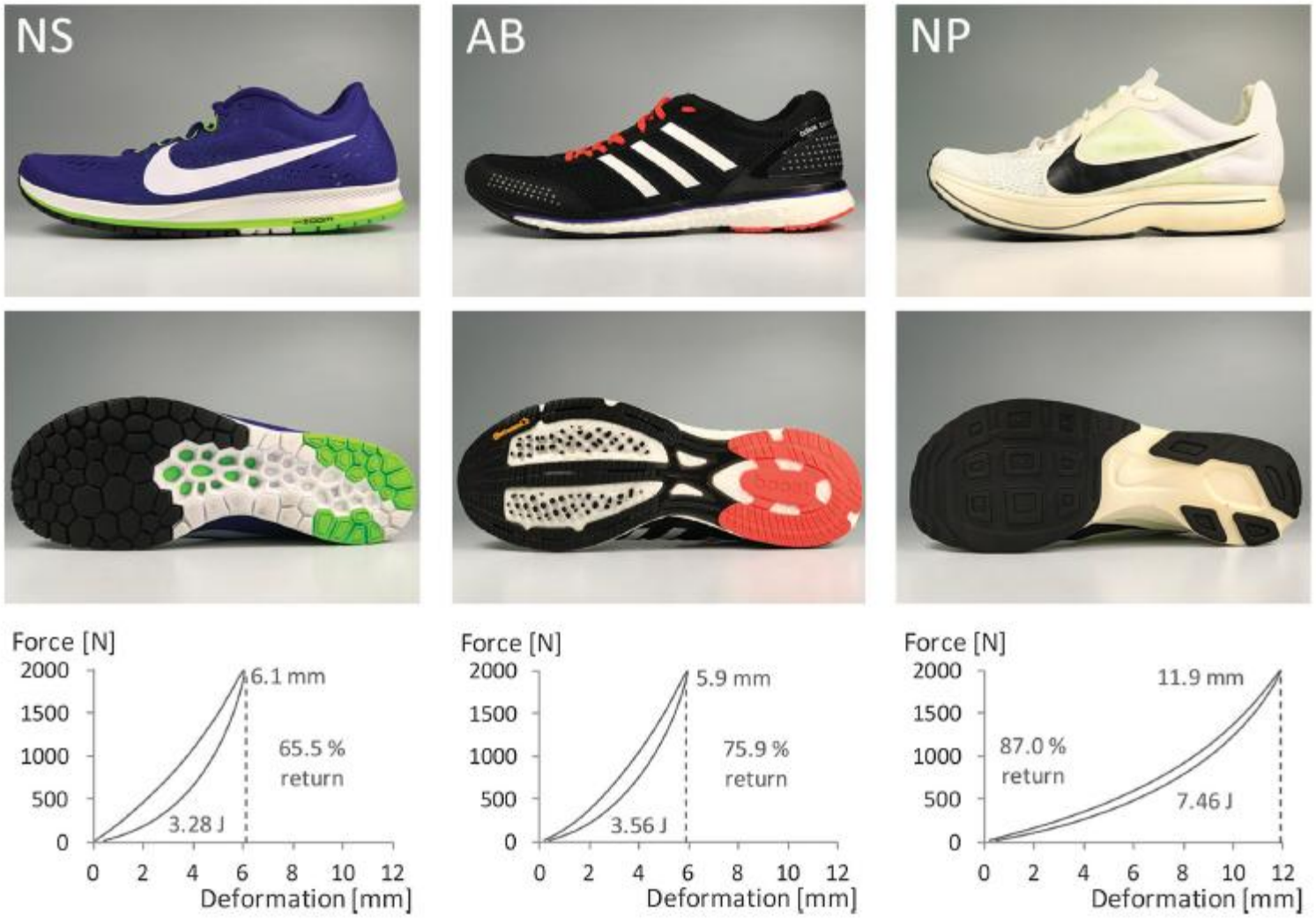
- Energy (work) is area under force vs. position curve
  - Energy stored during loading
  - Energy returned during unloading
- Parameters of interest
  - Absolute energy return (AER) in joules (J)
  - Relative (%) energy return (RER)







Cf approx. 70%; Stefanyshyn & Nigg. 2000. Sportverl Sportschad

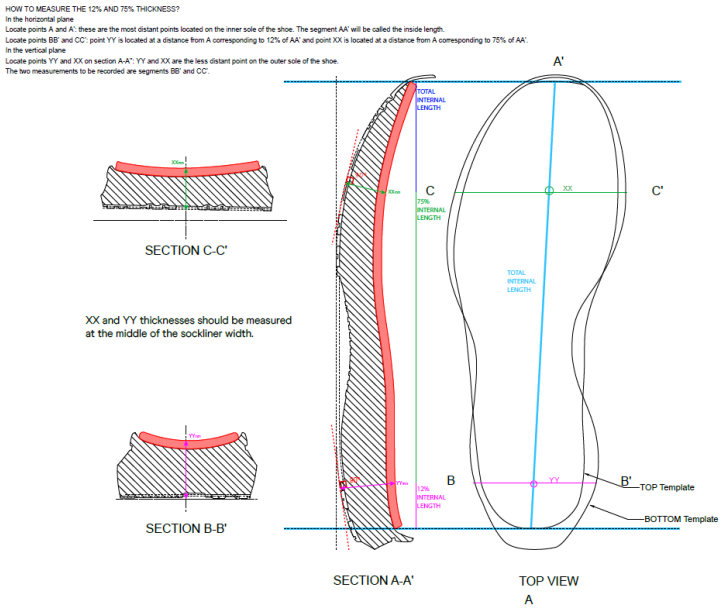


Hoogkamer *et al*. Sports Medicine, 2018

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Tests for energy return

Forefoot compression



Heel compression



Tensile bending stiffness



Hallström *et al*., under review, Footwear Science

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Methods

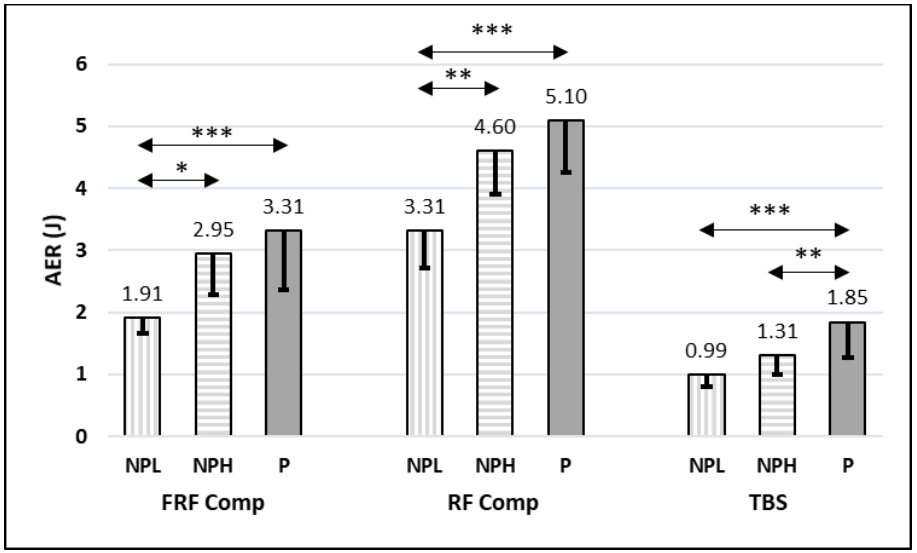
- 56 shoe models tested, AER och RER
- Shoes grouped into three groups based upon WA rule indicators: stack height at the heel and presence of rigid elements
    1. No plate, height ≤ 29 mm
    2. No plate, height ≥ 29 mm
    3. With plate

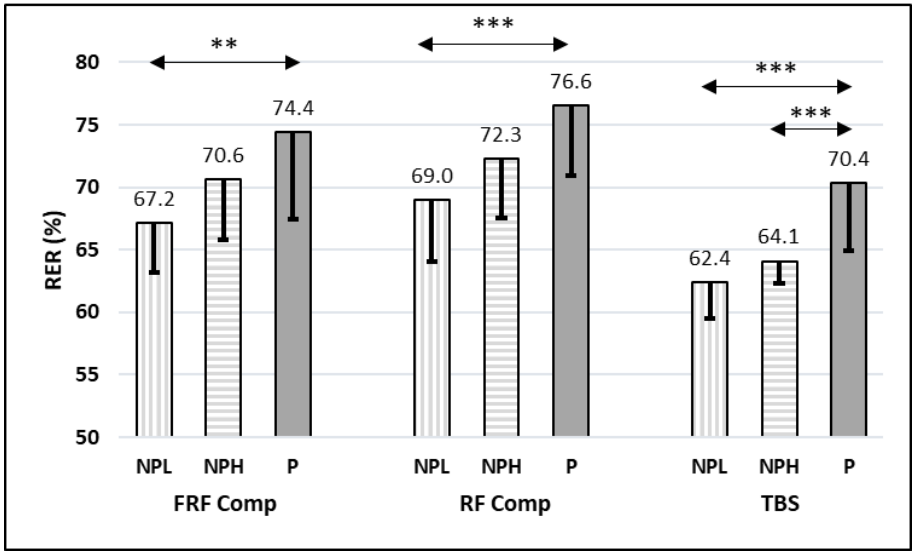
Arndt *et al*., almost submitted somewhere

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Results AFT vs other road shoes

Energy return in three shoe categories





Arndt *et al*., almost submitted somewhere

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Results

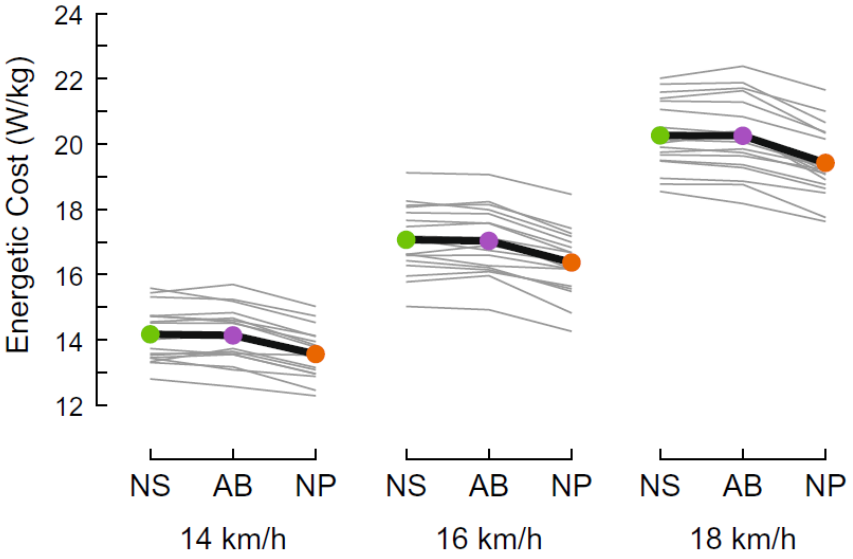
- Higher stack heights improved both AER and RER
- Including a rigid element further increased AER and RER by approx. 5-6% in compression tests
- Stack height had no effect on the tensile bending test (TBS) but this was affected by a rigid element
- RER in the TBS increased by approx. 10% with compared to without a rigid element

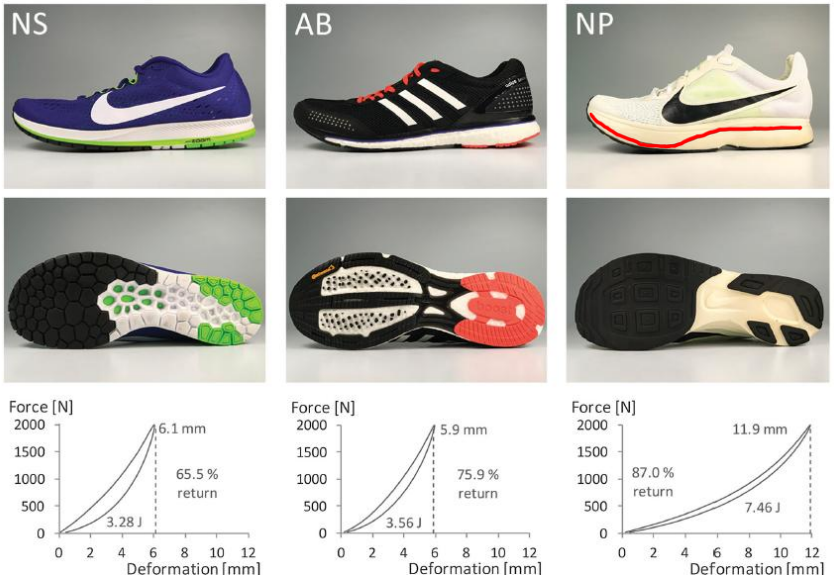


Arndt *et al*., almost submitted somewhere

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Influence of footwear properties on running economy





Running economy improved by 4%

= 4 mins 48 s for a 2 hour marathon

Hoogkamer *et al*., 2018. Sports Medicine

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Influence of footwear properties on running economy



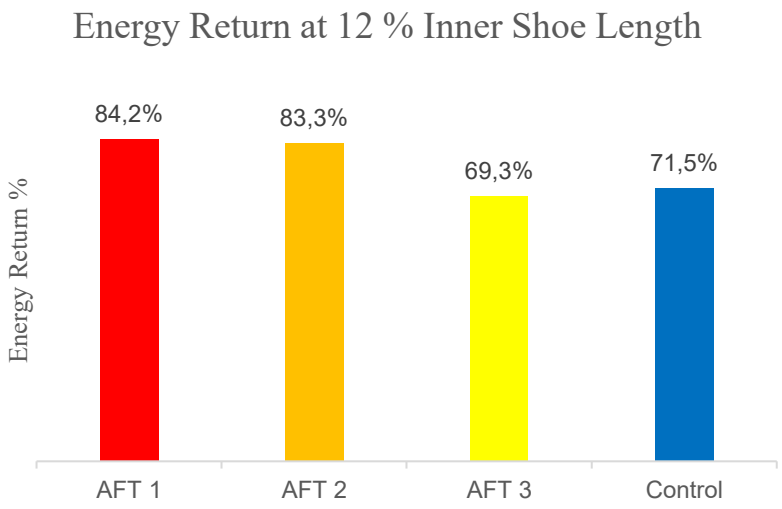
CENTRUM FÖR IDROTTSFORSKNING

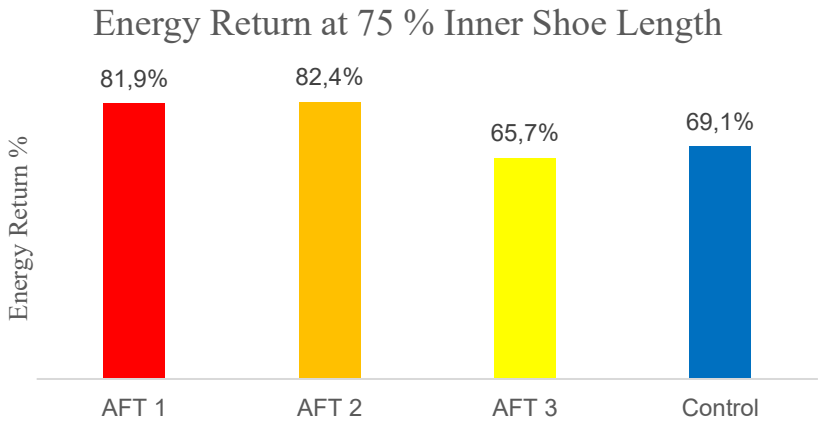
Vetenskapsrådet

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Shoes

- Three AFT shoes and one non-AFT shoe
  - Two of the top performing (Energy Return) AFT shoes and one AFT shoe with similar values as the non-AFT shoe





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Methods

- Visit 1: VO2peak test to determine VO2peak velocity
- Visit 2: Running economy tests in four shoes (3 AFT and 1 non-AFT)
  - Shoe order randomised
  - 4 minutes running at 60%, 70% and 80% of VO2peak velocity



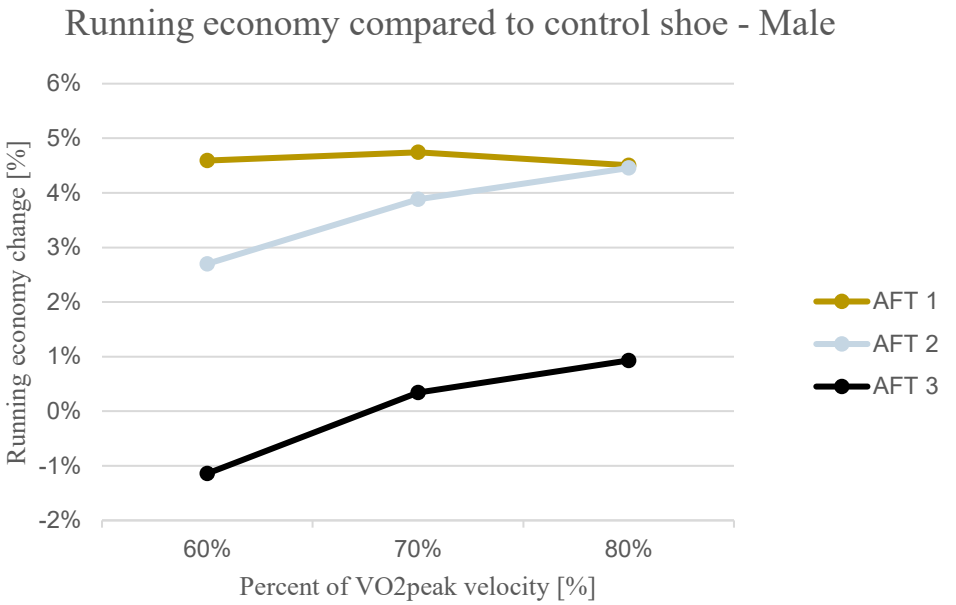
Torniainen *et al*., under review

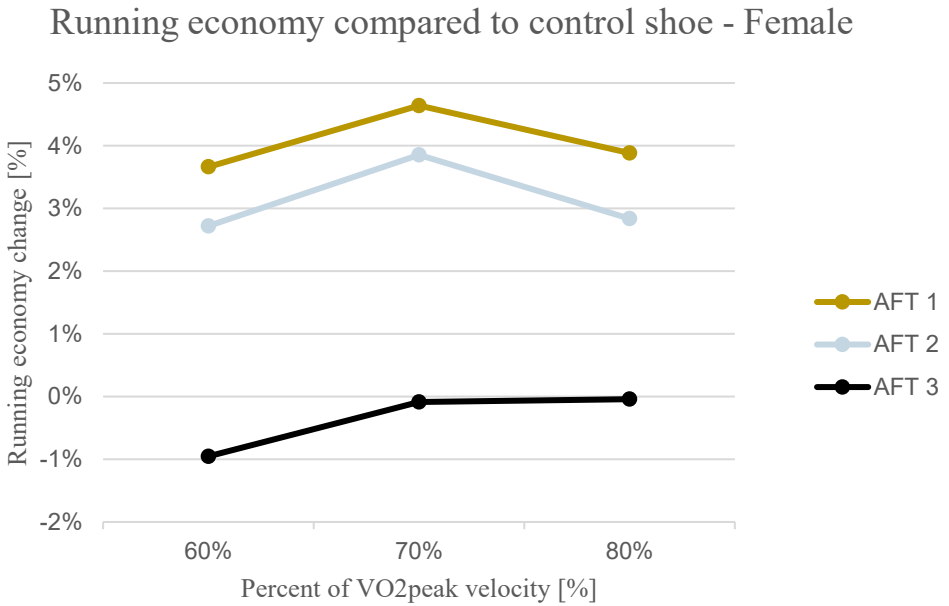
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Results

- Running economy improvements compared to non-AFT shoe at all three intensities

- The high energy return AFT shoes outperform the low energy return AFT shoe for both male and female runners

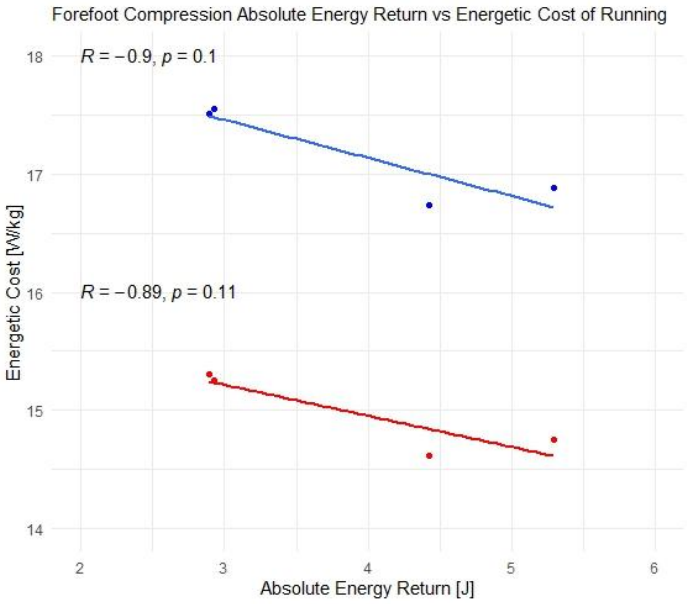


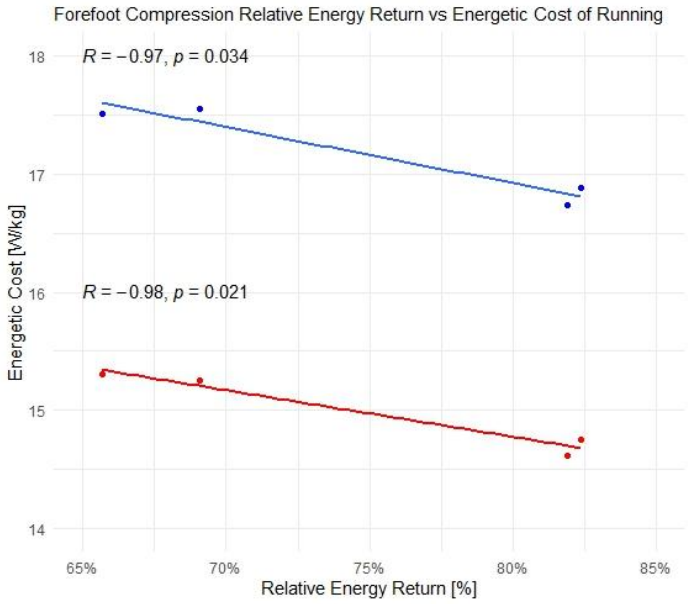


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Results: forefoot compression







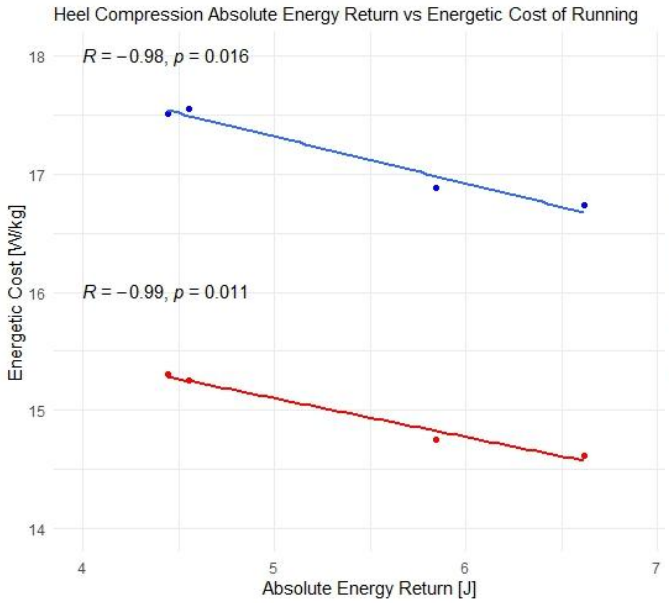
better

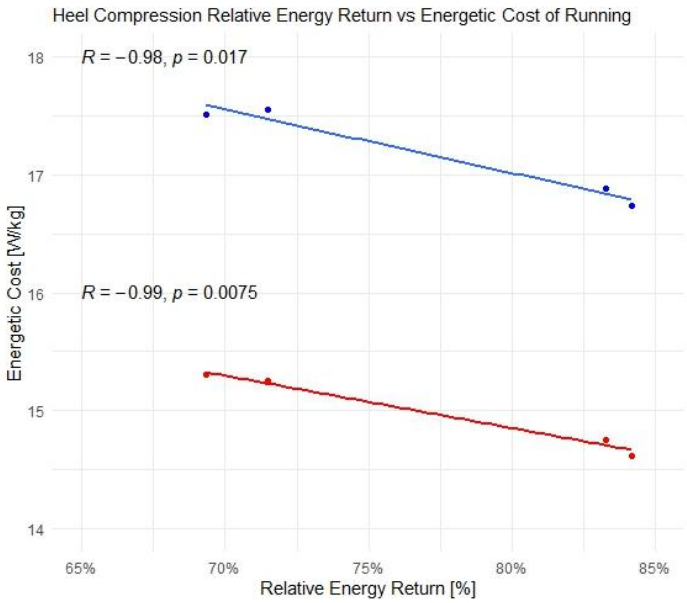
better

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Results: heel compression



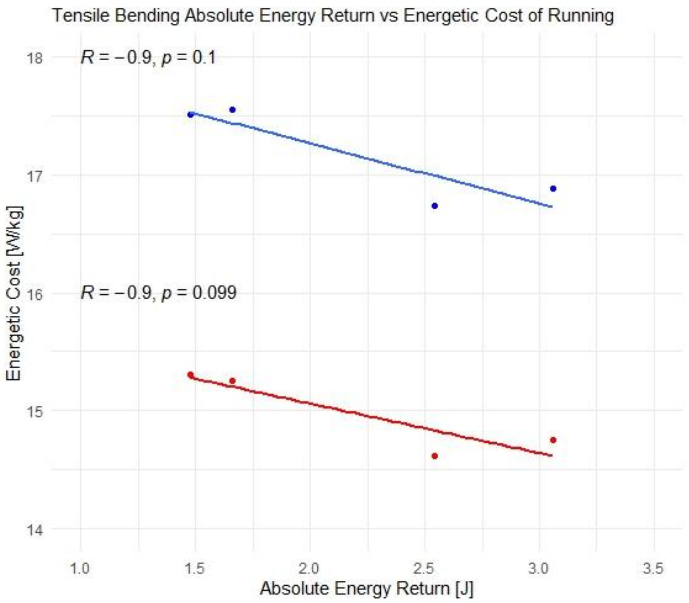


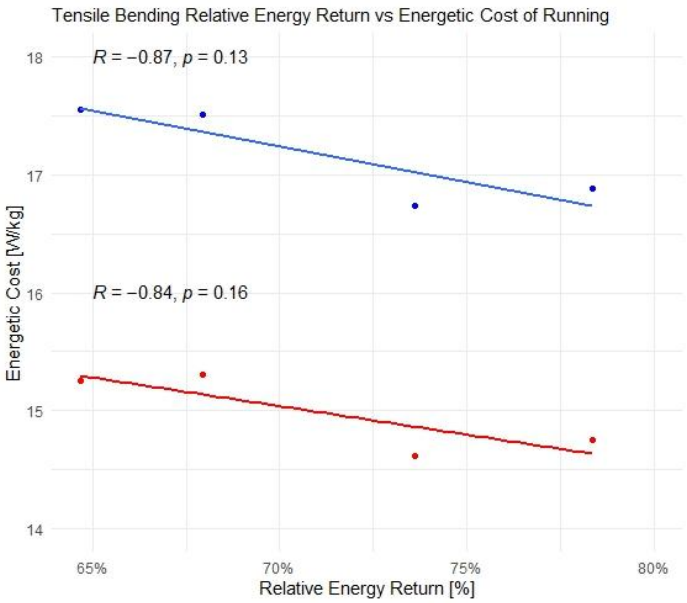


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Results: tensile bending stiffness







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Results

- Significant correlations between energy return parameters and running economy
- For male and female runners
- Unknown previously
- Consequences for shoe regulation?



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What now?

- Publication of strength and anthropometric individual characteristics
- Biomechanical risk factors for injury in AFT?
- Longitudinal study of injuries
- Exploring energy return: feet – shoes - prostheses
- Elite sustainability: injury prevention and performance

CENTRUM FÖR IDROTTSFORSKNING

Vetenskapsrådet

KK-stiftelsen

SVENSK FRIIDROTT

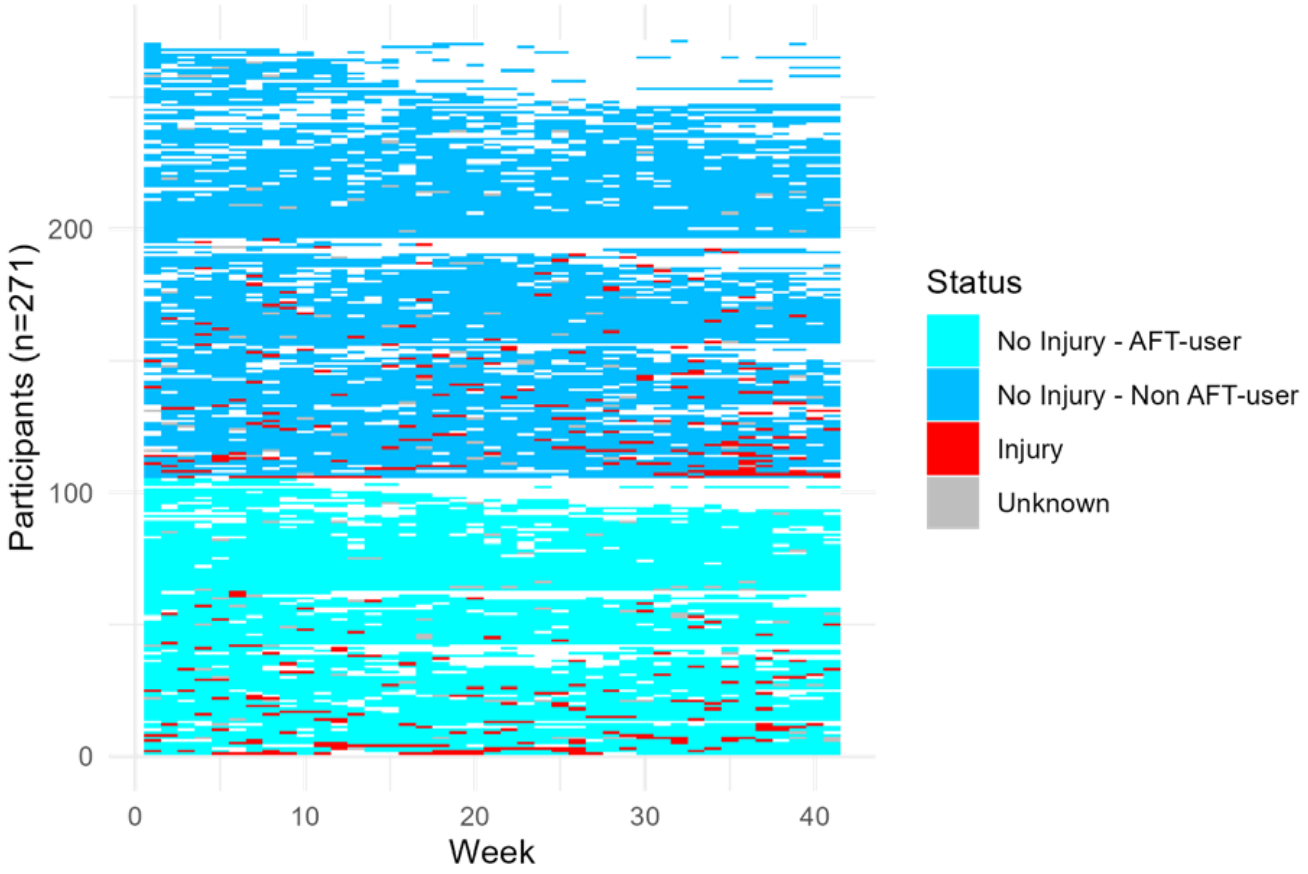
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Implications for injury / injury risks?

- Intuitively: longer lever arm, stack height, instability, foot passiveness
- No scientific studies to date showing a clear relationship between AFT and injury risk
- Anectdotal evidence
- Biomechanical risk factors? PhD Project, GIH / Deutsche Sporthochschule Köln
- One year longitudinal follow-up of recreational runners, GIH / Linköping University

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Preliminary data: longitudinal follow-up



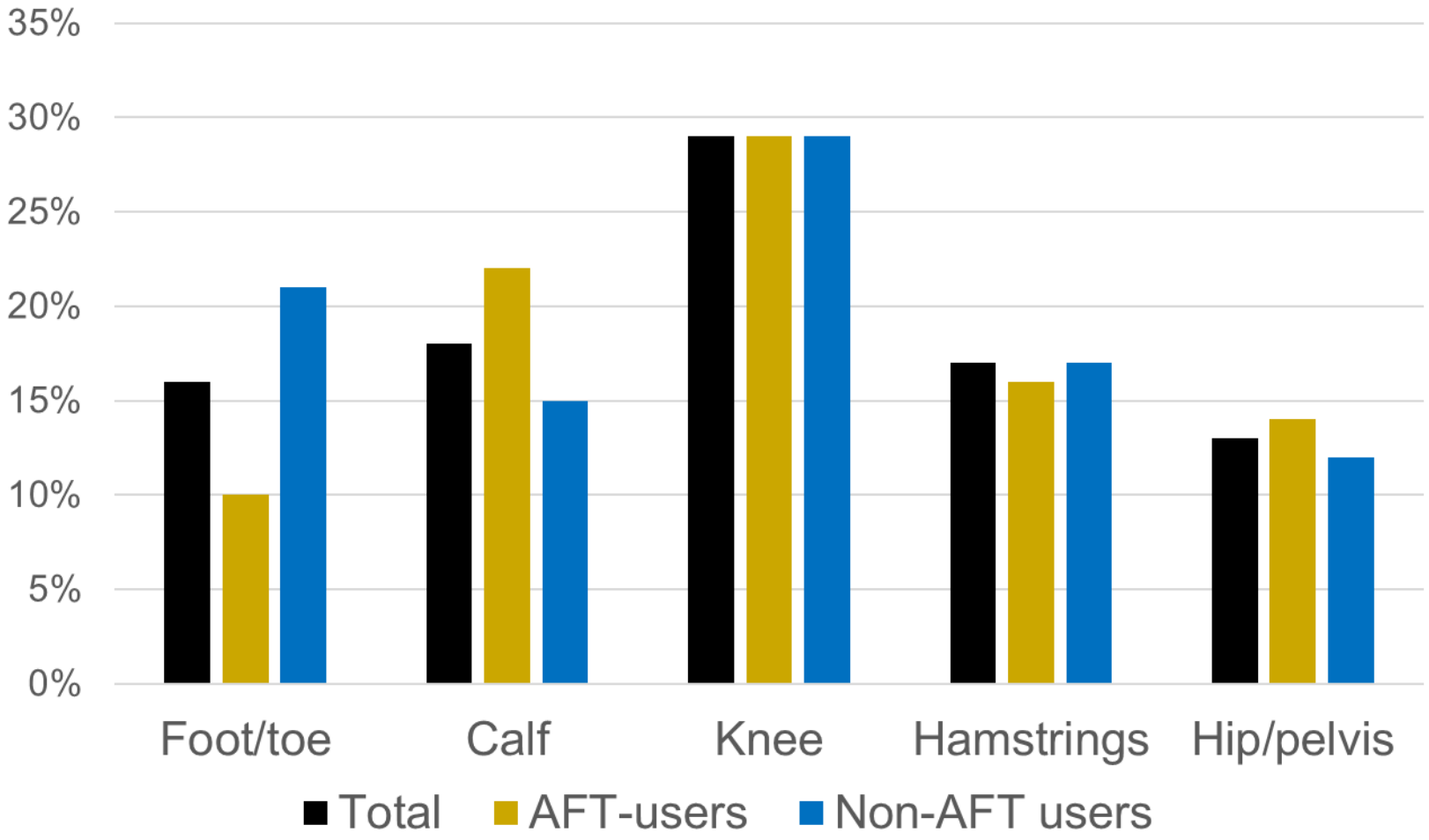
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Preliminary data: longitudinal follow-up

	Total	AFT-users	Non-AFT users	Rate Ratio	P
Incidence RRI	8.4	7.9	8.8	0.9	ns
Males	7.5	7.2	7.8	0.9	ns
Females	9.8	9.7	9.9	1.0	ns
Prevalence	1.8	2.1	1.6	1.3	ns
Males	1.7	2.2	1.3	1.7	0.043
Females	1.9	1.9	1.9	1.9	NS

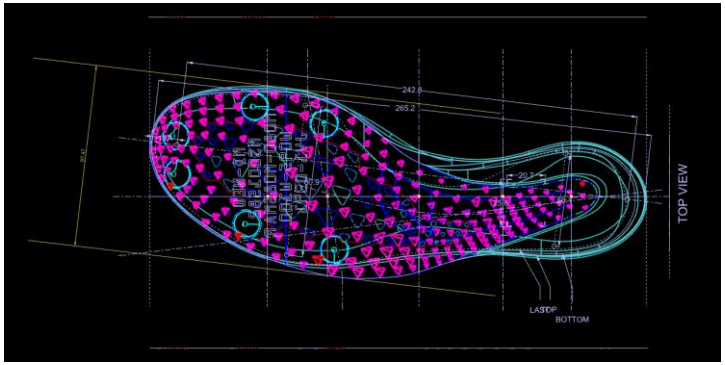
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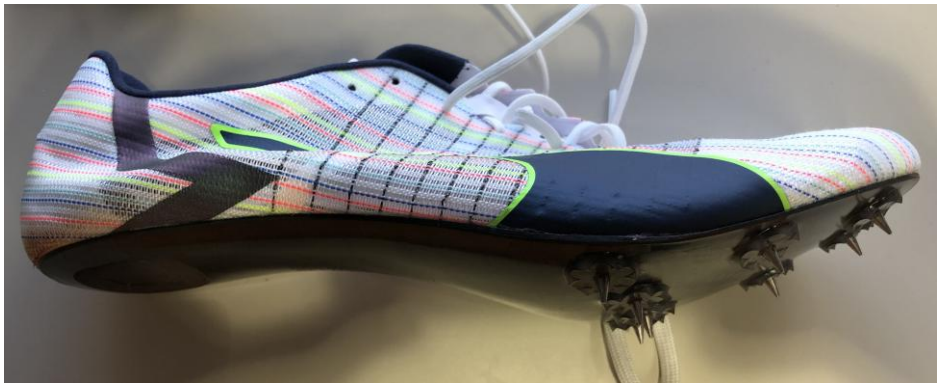
Preliminary data: longitudinal follow-up



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Performance in AFT spikes









World Athletics spike rules

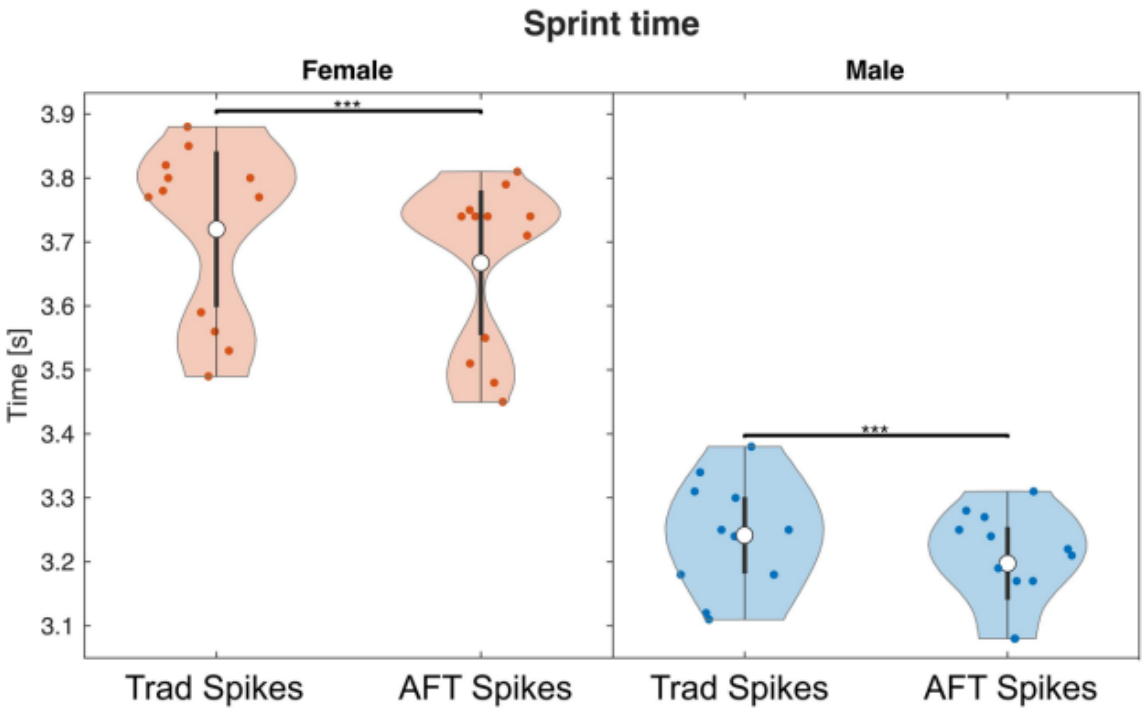
- I.e. for track and field events
- Apart from standard spike regulations
- Height 20 mm (modified in 2024)

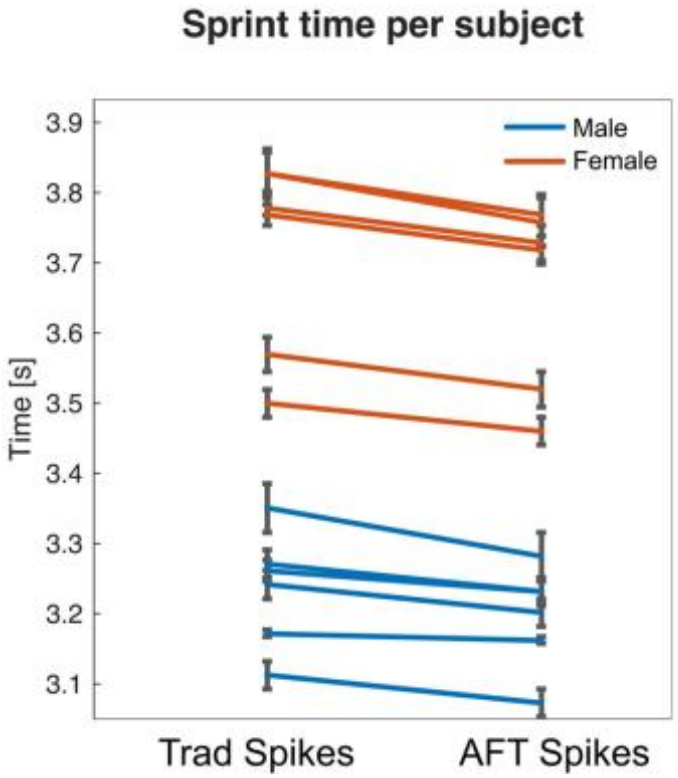
- Field events not higher at 75% than 12%
- Additional rigid element ”solely for the insertion of spikes”



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Performance in AFT spikes





Gerlach *et al*. 2025, Footwear Science

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Performance in AFT spikes

- Jumping?



- Throwing?



- Relevance to results across completely different events?
- Relevance of running economy?

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Ruth Chepngetich marathon
World record 2:09:56 s



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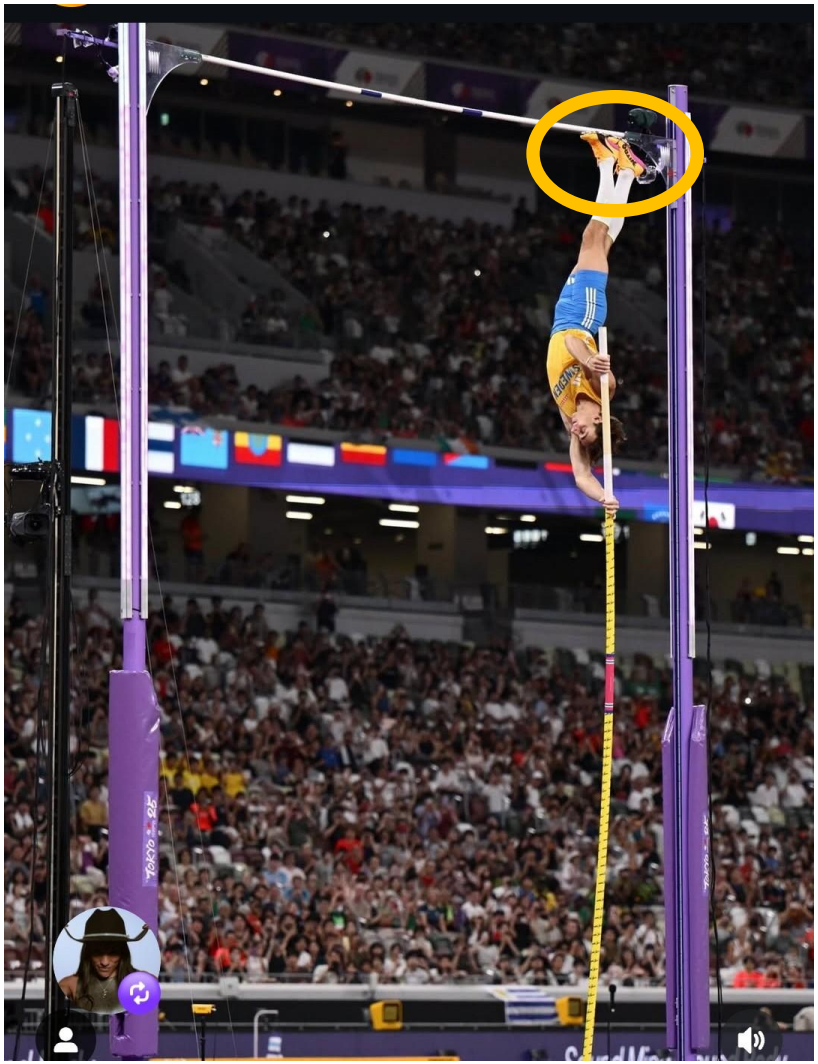
Sydney McLaughlin Levrone 400 m hurdles
World record 50.37 s



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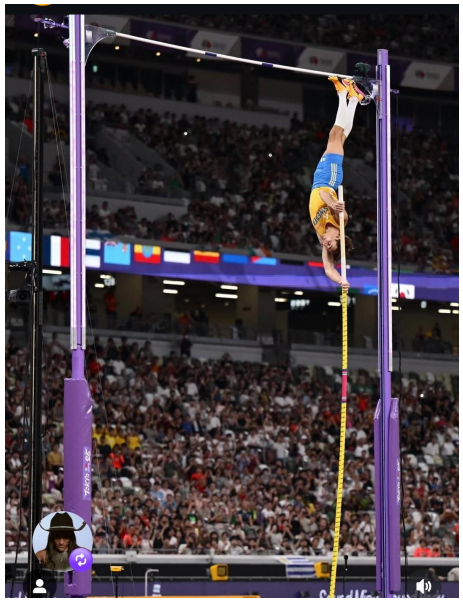


Armand Duplantis polevault
World record 6.30 m



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Spectrum of AFT influence on results in different athletic events







least

most

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What should/needs to be regulated?

To facilitate evaluation of new technology and prevent negative consequences for athletes and manufacturers, a clearly defined limit of allowed performance enhancement from the technology should be provided by the federation.

Technological developments in sports – A question of fairness and integrity? Masters thesis GIH 2020 Frida Williamson.

Arndt *et al*, under review, Journal of Philosophy of Sportc

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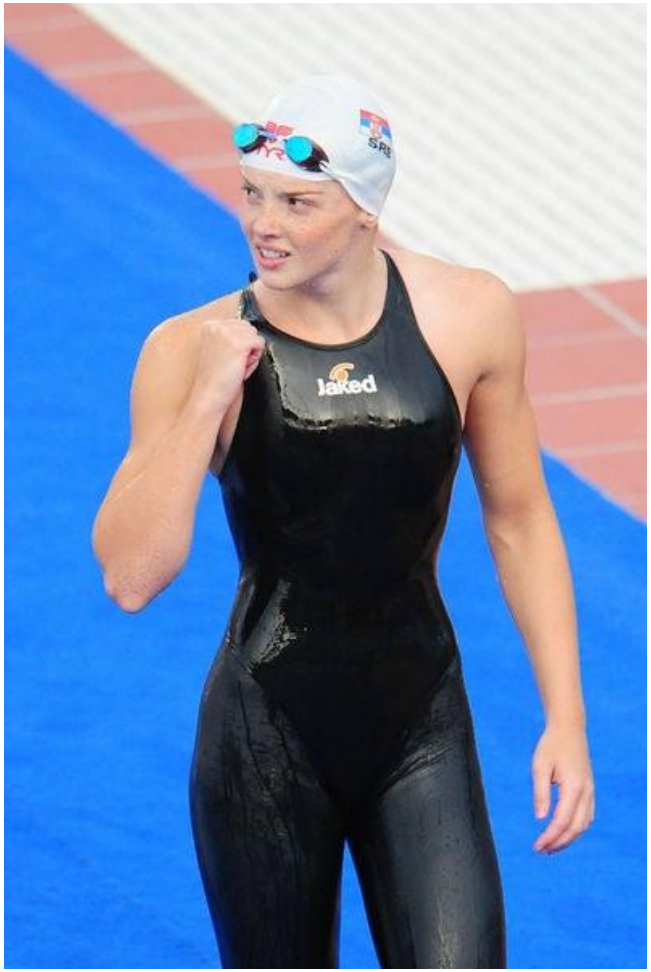
You can swim naked

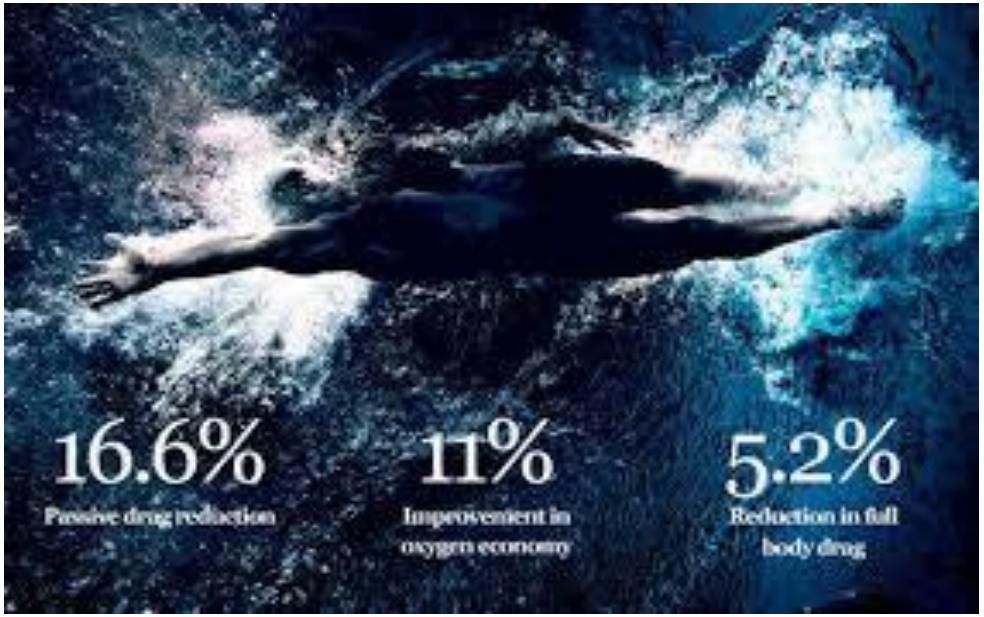


Arndt *et al*, under review, Journal of Philosophy of Sport

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You can swim naked

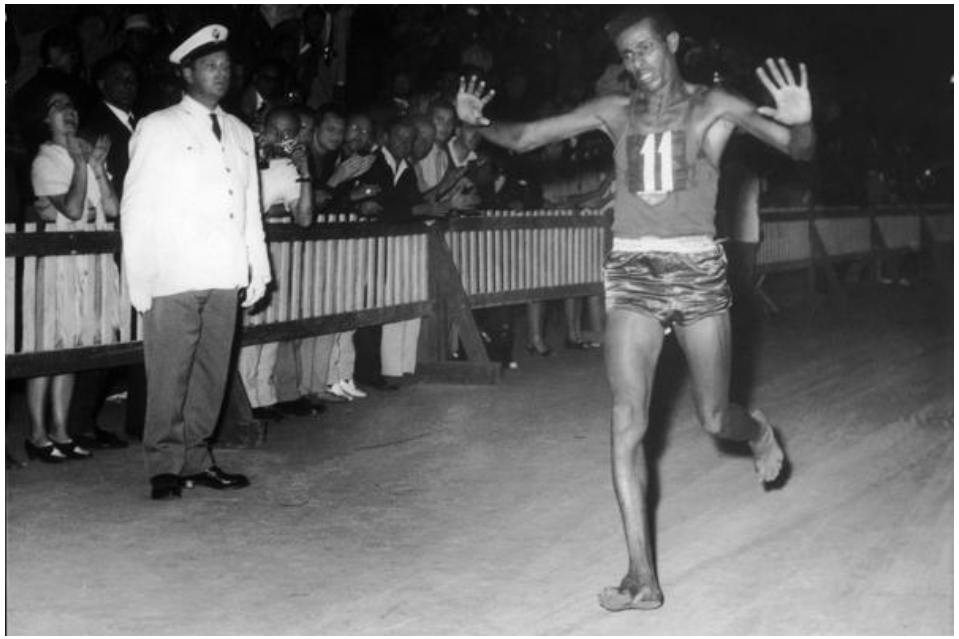




Arndt *et al*, under review, Journal of Philosophy of Sport

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You can run barefoot





EXCLUSIVE: Call for 'investigation' as multibillion-dollar, Nike-led 'super shoe' era rocks athletics

Arndt *et al*, under review, Journal of Philosophy of Sport

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Thank you